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厦門大學馬來西亞分校

THESIS PROPOSAL

AUDIO PRE-PROCESSING AND COMPARATIVE TEXT CLASSIFICATION FOR SPEECH-BASED MEDICAL SYMPTOM UNDERSTANDING

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JUNE 2026

CHAPTER 1
INTRODUCTION

1.1 Introduction

Nowadays, medical chatbot systems play an important role in healthcare because they can support patient communication, self-management and early guidance before users reach a clinician (Lizée et al., 2025). Recent reviews show that conversational agents are increasingly explored for healthcare communication and chronic disease support, but reliability, safety, and user trust remain important concerns (Shekar et al., 2024). In addition, recent medical chatbot research notes that current systems still struggle with imprecise understanding of medical language, jargon, and personalized feedback, which limits their practical usefulness (Nipu et al., 2024).

To improve user interaction, speech input can be used as the front-end of the system so that users can describe symptoms naturally (Adedeji et al., 2024). Whisper is a strong pretrained ASR baseline because it was trained on 680,000 hours of multilingual and multitask supervision, and it shows robust zero-shot performance under accents, background noise, and technical language (Banerjee et al., 2024). This makes Whisper suitable for speech-to-text conversion in a medical chatbot, especially when the system needs to handle real-world voice input (Radford et al., 2022).

However, speech recognition alone is not enough for medical understanding, because symptom descriptions are often short, informal, and expressed in different ways (Mottaghi et al., 2021). Traditional approaches such as TF-IDF with Logistic Regression remain widely used due to their computational efficiency and strong baseline performance on sparse text data (Ramos, 2003). Transformer-based models such as BERT have also demonstrated strong performance on general NLP classification tasks (Devlin et al., 2019), while domain-specific models such as pretrained model BioBERT including PubMed abstracts and PubMed Central articles, have been shown to better capture medical terminology and outperform general BERT on biomedical text mining tasks (Lee et al., 2020). This supports the need for a comparative evaluation of text classification models to

identify the most suitable approach for symptom-based disease classification after speech-to-text conversion.

Based on these issues, this thesis proposes a speech-based medical chatbot that combines pretrained ASR, custom text classification, and guided response generation in a user-friendly prototype. The system is designed to convert user speech into text, identify symptom-related content, and return practical guidance rather than medical diagnosis. Recent healthcare chatbot studies also emphasize that chatbot systems should be usable, trustworthy, and effective from the user's perspective, which supports the need for a practical and user-friendly prototype (Bhattacharya & Pissurlenkar, 2023).

1.2 Problem Background/Problem Statement

Although pretrained ASR models such as Whisper have significantly improved speech-to-text performance, speech recognition errors can still affect downstream language understanding tasks (Peng et al., 2023). This issue is important in medical chatbot systems because symptom descriptions are often short, informal, and difficult to interpret correctly (Mottaghi et al., 2021). Medical text understanding is also challenging because symptom labels can be multi-label and long-tailed, while healthcare chatbots still face limitations in usability, trust, and reliable communication (Mottaghi et al., 2021).

In my own test run, the sentence "I have a headache every day in the morning" was transcribed as "I have a haydack every day, in the morning." This shows that even a small transcription error can change a medical term and affect the accuracy of symptom classification and chatbot response generation (Shapira et al., 2025). If the text cannot be correctly recognized by ASR, the system might wrongly interpret the user and provide a wrong result (Shapira et al., 2025).

1.3 Aims and Objectives

The purpose of this project is to create a medical chatbot with speech-to-text capabilities and post-process the extracted text to enhance its quality for subsequent use. This study compares 3 different medical text classification models: TF-IDF with Logistic Regression,

fine-tuned BERT and fine-tuned BioBERT for 24 medical classes, to offer the user a friendly and useful medical advice. This project will focus on the symptoms of headache, fever, cough, sore throat, stomach pain, nausea, dizziness and mild skin complaints. The system is designed to help with the initial symptoms of a problem, not a detailed diagnosis

The objectives of this study are:

a) To perform post-processing of speech to text conversion for enhancing the quality of the transcript.

b) To compare three models, TF-IDF with Logistic Regression, fine-tuned BERT and fine-tuned BioBERT for 24 classes of disease classification and choose the best performing model for the 24 disease classes.

c) To create a prototype that is easy to use and enable the user to communicate with the medical chatbot through speech and text input.

1.4 Scope of the Project

The scope of this project is limited to the development of the prototype of a speech-enabled medical chatbot. The system focuses on symptom screening and basic medical guidance rather than full clinical diagnosis. The scope of the project is as follows:

a) To convert user speech into text using a pretrained Whisper model.

b) To perform post-processing on the transcribed text to further enhance text quality.

c) To categorize the text into symptom-related categories with a trained text classification model.

d) To provide simple medical advice or answers depending on the anticipated class of disease.

e) To create a user-friendly prototype enabling users to interact orally and in text.

f) To add an optional feature of image input and recommendations for nearby clinic/hospital in case of urgency.

1.5 Expected Contribution of the Project

This project is expected to contribute a practical framework for speech-based medical interaction. The expected contributions are as follows:

a) A speech-to-text medical chatbot pipeline that combines pretrained ASR with custom text classification.
b) Post-process algorithm after the speech-to-text conversion.
c) Comparative analysis of TF-IDF, fine-tuned BERT and fine-tuned BioBERT for 24-class disease classification and then choosing the best model for the Chatbot.
d) A user-friendly chatbot prototype that supports basic medical guidance.

CHAPTER 2
LITERATURE REVIEW

2.1 Introduction

This chapter reviews existing literature related to the three main components of the proposed system which are automatic speech recognition, medical text classification, and response generation. Section 2.2 reviews the development of ASR systems and compares pretrained models including Whisper, wav2vec2, and Conformer to justify the selection of Whisper as the speech-to-text backbone. Section 2.3 covers medical symptom understanding as a text classification task and reviews three models evaluated in this study which are TF-IDF with Logistic Regression, BERT, and BioBERT in terms of their characteristics and suitability for disease classification. Section 2.4 reviews semantic similarity techniques and response generation approaches relevant to medical chatbot systems. Section 2.5 discusses the challenges of integrating ASR with downstream language understanding modules in a medical chatbot pipeline. Section 2.6 identifies the research gap that motivates this study, and Section 2.7 summarizes the key findings from the literature.

2.2 Speech-to-Text and Automated Speech Recognition

ASR has progressed from GMM-HMM systems, DNN-HMM systems to end-to-end neural systems such as CTC, attention-based encoder-decoder systems, RNN-T, Transformer-based systems, and Conformer-based systems (Nayeem et al., 2025). At the same time, self-supervised and weakly supervised models have further reduced the dependence on large labelled speech datasets (Baevski et al., 2020).

This advancement has enhanced the capabilities of ASR and rendered it more practical for real-world speech transcript (Nayeem et al., 2025). However, could see that the literature also shows that model performance still affected by the speech domain, the number of labelled data, and the type of acoustic variation in the input (Hsu et al., 2021).

2.2.1 Comparison of Pretrained Automatic Speech Recognition Models

Recent ASR research has focused on pretrained models such as Whisper, wav2vec2, and Conformer-based architectures developed by major research groups including OpenAI, Meta AI, and Google Research (Gulati et al., 2020; Jain et al., n.d., 2017). Whisper was developed by OpenAI as a weakly supervised multilingual ASR model, wav2vec2 was introduced by Meta AI using self-supervised speech representation learning, while Conformer was proposed by Google Research by combining convolution and Transformer attention mechanisms (Gulati et al., 2020; Jain et al., n.d., 2017).

A comparative study by Barcovschi et al. (2023) evaluated Conformer-Transducer, Whisper, and wav2vec2 on the same child speech datasets. Their results showed that wav2vec2 achieved the most consistent performance improvements overall, while Whisper remained a strong pretrained baseline with robust generalization across different speech conditions shown in Table 2.2.1 (Barcovschi Student et al., n.d.).

In addition, studies on Whisper adaptation reported that finetuning Whisper improves recognition performance in domain-specific speech tasks, although wav2vec2 may outperform Whisper under certain specialized datasets. Therefore, Whisper is selected as the pretrained ASR baseline in this study because it is robust, widely supported, and practical for real-world speech-to-text conversion.

Model	Characteristic	Best reported WER in the paper (MyST / PFSTAR / CMU) Speed	Suitability for Medical Chatbot
wav2vec2	Self-supervised speech representation model; data-efficient and strong after fine-tuning.	**7.42 / 2.99 / 14.18** (large, fine-tuned on MyST+PFSTAR)	Best overall accuracy in this benchmark, so it is useful as a strong comparison baseline.

Conformer-Transducer	Combines CNN and Transformer blocks; strong local and global speech modeling, but heavier training/decoding cost.	**13.61 / 4.30 / 21.21** (XLarge, fine-tuned on MyST+PFSTAR)	Strong baseline, but more computationally demanding and less consistent on harder datasets.
Whisper	Large-scale weakly supervised multilingual model; strong zero-shot robustness and no task-specific fine-tuning requirement.	**13.34 / 4.17 / 17.11** (Large-V2, fine-tuned on MyST+PFSTAR)	Good pretrained baseline for robust transcription, but not the lowest WER in this paper.

Table 2.1: Comparison of Selected Pretrained ASR Models Based on Barcovschi et al. (2023)

2.3 Medical Symptom Understanding

Medical symptom understanding is formulated as a text classification task, where patient-generated text is mapped to a corresponding disease class. Studies have shown that approaches ranging from traditional feature extraction to transformer-based models can effectively support this task (Kowsari et al., 2019; Mottaghi et al., 2021). In this study, three models are evaluated and compared: TF-IDF with Logistic Regression, fine-tuned BERT, and fine-tuned BioBERT.

2.3.1 TF-IDF with Logistic Regression

TF-IDF assigns numerical weights to terms based on their frequency within a document relative to the entire corpus, effectively highlighting symptom-relevant keywords in patient text (Ramos, 2003). When combined with Logistic Regression, this approach has demonstrated competitive classification performance on medical text and has outperformed several machine learning models in healthcare text classification tasks (Kalra et al., 2019; Singh et al., 2025).

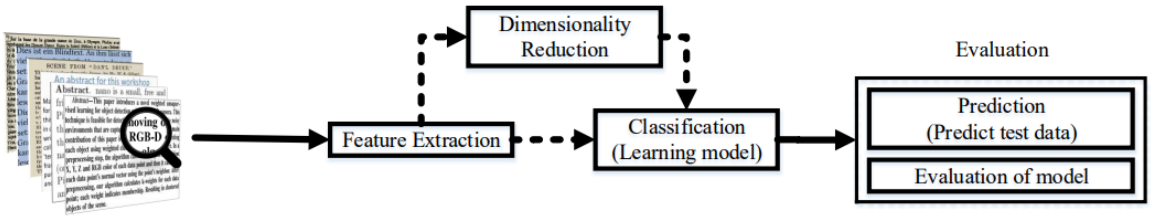


Figure 2.1, Text classification pipeline (Kowsari et al., 2019)

2.3.2 BERT for Medical Text Classification

BERT is a pretrained transformer model that captures deep bidirectional contextual representations from large general text corpora and can be fine-tuned for downstream classification tasks (Devlin et al., 2019). Although fine-tuned BERT has been applied to medical text classification, Li et al. (2022) found that on disease classification tasks with limited labelled data, simple linear classifiers outperformed BERT when training samples per class were very few. Given that this study's dataset contains approximately 50 samples per class, BERT is included as a comparison model to assess whether contextual representations provide any advantage at this data scale.

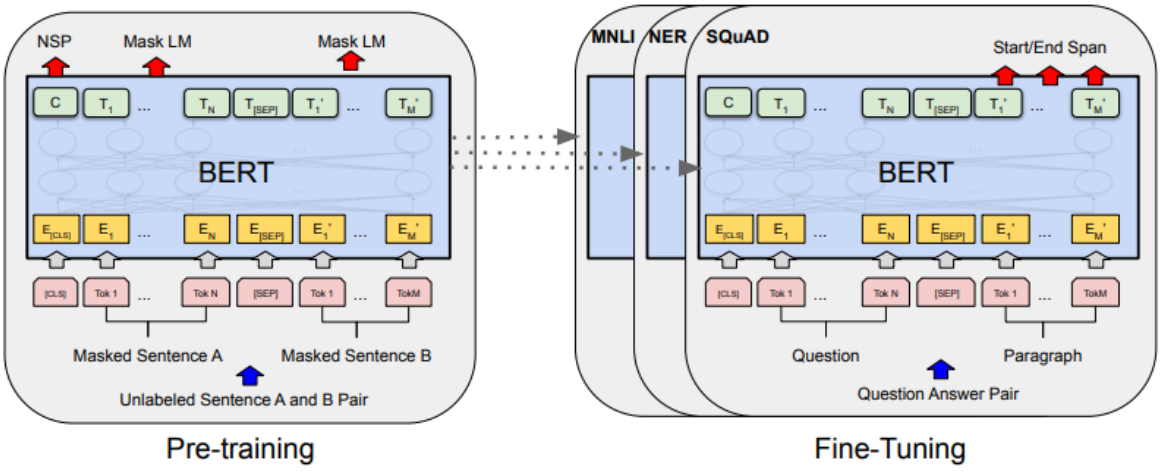


Figure 2.2, BERT pre-training and fine-tuning procedures for downstream classification tasks (Devlin et al., 2019)

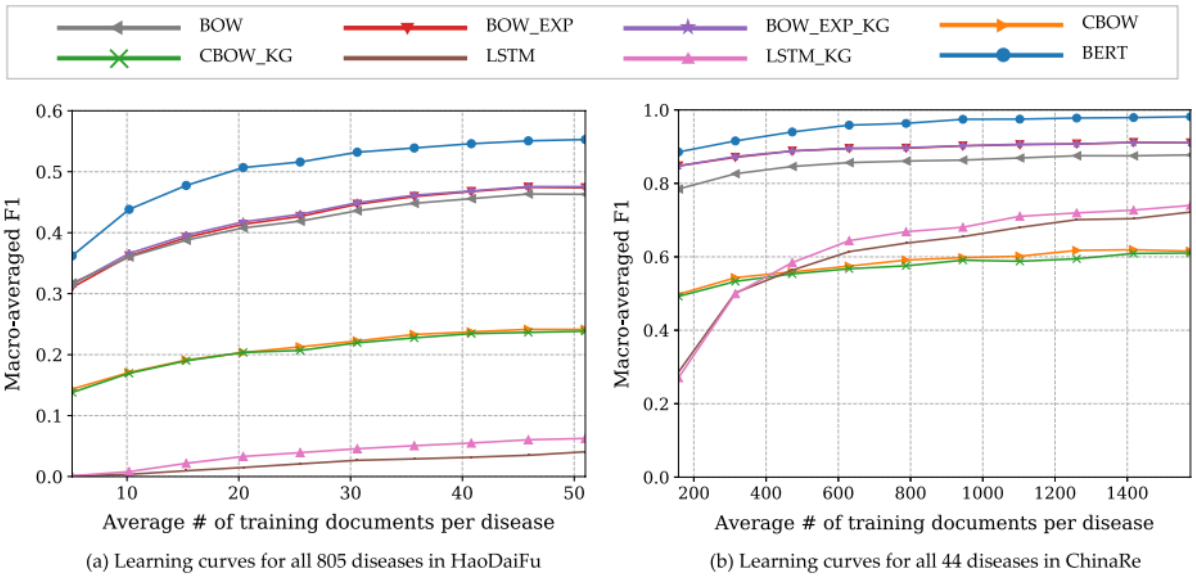


Figure 2.3, Learning curves of BERT and baseline classifiers on disease classification tasks, showing BERT is outperformed by linear classifiers under extreme data scarcity (Li et al., 2022)

2.3.3 BioBERT for Medical Text Classification

BioBERT extends BERT by further pretraining on PubMed abstracts and PubMed Central full-text articles, enabling it to better represent biomedical vocabulary and medical terminology not well captured by general-domain pretraining (Lee et al., 2020). BioBERT has been shown to outperform general BERT across biomedical NLP tasks, and in disease classification from symptom text specifically, Hassan et al. (2024) found that its domain-specific pretraining yielded superior performance. BioBERT is therefore included as the third model to evaluate whether biomedical pretraining improves classification.

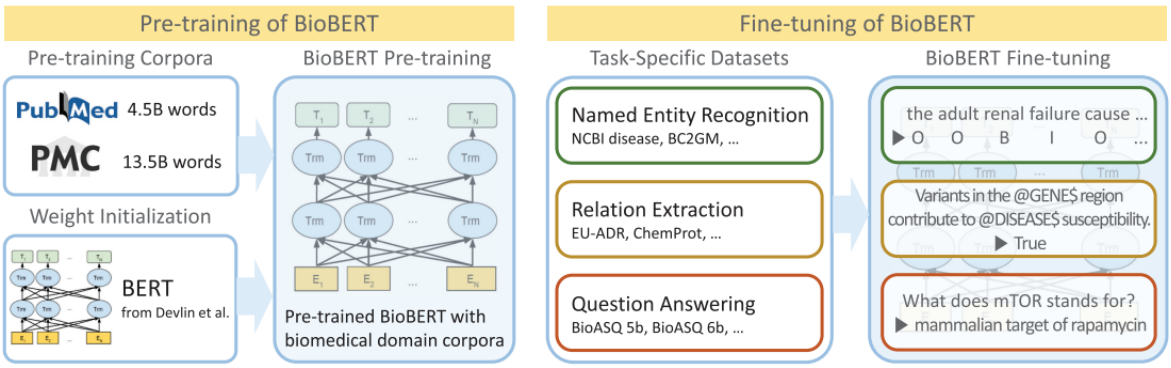


Figure 2.4, Pre-training and fine-tuning process of BioBERT using PubMed and PubMed Central corpora (Lee et al., 2020)

2.4 Semantic Similarity and Response Generation

Although this study uses rule-based response generation, semantic similarity is reviewed here as relevant prior work in medical chatbot response retrieval.

2.4.1 Semantic Similarity in Medical Response Retrieval

Semantic similarity is used to match user input with the most relevant medical response when the wording is different but the meaning is similar (Reimers & Gurevych, 2019). Sentence-BERT is suitable for this task because it generates sentence embeddings that can be compared efficiently using cosine similarity (Reimers & Gurevych, 2019).

In retrieval-based chatbot systems, semantic search is important for selecting candidate responses from a response pool efficiently. A neural matching study on chatbot candidate retrieval showed that query-response interaction modelling can improve semantic search for chatbot response selection (Shi et al., 2022).

In medical dialogue, semantic similarity has also been shown to improve question understanding. A virtual patient study reported that combining rules with semantic similarity achieved a higher F1-score than rules alone, Figure 2.4.1 showing that semantic matching can improve medical query interpretation (Laleye et al., 2019).

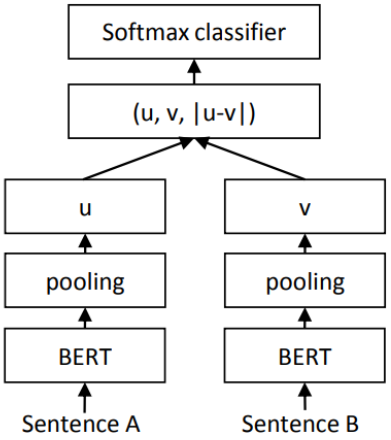


Figure 2.5, Siamese BERT network architecture used in Sentence-BERT for computing semantic similarity between sentence pairs (Reimers & Gurevych, 2019)

2.4.2 Response Generation and Safety Handling

Semantic Medical chatbot responses should provide guidance rather than diagnosis. A real-world evaluation of physician-supervised conversational medical AI found that carefully implemented systems can improve patient experience while maintaining safety under oversight (Lizée et al., 2025).

However, users may overthrust AI-generated medical advice, even when the response is inaccurate. Study found that non-experts rated low-accuracy medical responses as trustworthy and were willing to follow them (Shekar et al., 2024). Therefore, response generation should include safety filtering, urgency detection, and referral advice for high-risk cases. Red-teaming studies of public medical chatbots also reported unsafe answers in a substantial number of cases, which supports the need for cautious medical response generation (Draelos et al., n.d.).

2.4.3 Prototype-Oriented Medical Interaction

A medical chatbot prototype should provide clear, simple, and safe interaction for users. Recent healthcare chatbot studies emphasize user-cantered design, usability, and trust as important requirements for practical deployment (Ganesan, n.d.). Real-world chatbot evaluations also show that accuracy, safety, and usability should be assessed together, not separately. PEACH was tested in a silent deployment using real-world clinical data, and user feedback was evaluated with the Technology Acceptance Model (Ke et al., n.d.). Therefore, the prototype in this study should focus on readable output, safe guidance, and easy interaction through speech or text input rather than on complex medical dialogue generation (Ganesan, n.d.).

Table 2.2, Summary of Previous Research on Medical Chatbot Systems

Author & Year	Techniques Used	Dataset	Results

Chakraborty et al. (2022)	NLP, deep feedforward multilayer perceptron (MLP), LSTM, named entity recognition	Custom JSON-format dataset covering infectious diseases, symptoms, treatment plans, and hospital information	Chatbot successfully predicted infectious diseases; reported competitive accuracy with lower time complexity compared to baseline approaches
Li et al. (2023)	LLaMA fine-tuned on medical domain knowledge with retrieval-augmented generation (ChatDoctor)	HealthCareMagic-100K (100,000 real patient-doctor conversations); iCliniq dataset	Outperformed GPT-3.5 on medical conversation tasks; improved accuracy in symptom understanding and disease-specific guidance
Yang et al. (2023)	Ensemble NLP deep learning model; multilingual support across 10 languages deployed on Telegram	2,728 training Q&As and 821 test Q&As on COVID-19 health queries	High overall and top-3 accuracy; strong AUC, precision, recall, and F1; outperformed enterprise-grade chatbot systems in multilingual evaluation
Nipu et al. (2024)	Evaluation of LLM-based chatbots (GPT-4, LLaMA,	Patient symptom complaint dataset	Significant variability in reliability across chatbots; non-expert

	Gemini) for medical reliability	(cross-chatbot evaluation)	users overtrusted low-accuracy responses; safety concerns identified
Hassan et al. (2024)	MCN-BERT (AdamP and AdamW optimisers), BiLSTM (Hyperopt)	Symptom2Disease (1,200 samples, 24 disease classes); Twitter ADR dataset (23,516 samples)	MCN-BERT (AdamP): 99.58% accuracy; MCN-BERT (AdamW): 98.33%; BiLSTM: 97.08% on disease classification from symptoms
Lizée et al. (2025)	Physician-supervised conversational AI (LLM-based), real-world clinical deployment	Real-world patient interactions from deployed clinical system	Patient experience improved under physician oversight; system demonstrated readiness for clinical practice with maintained safety

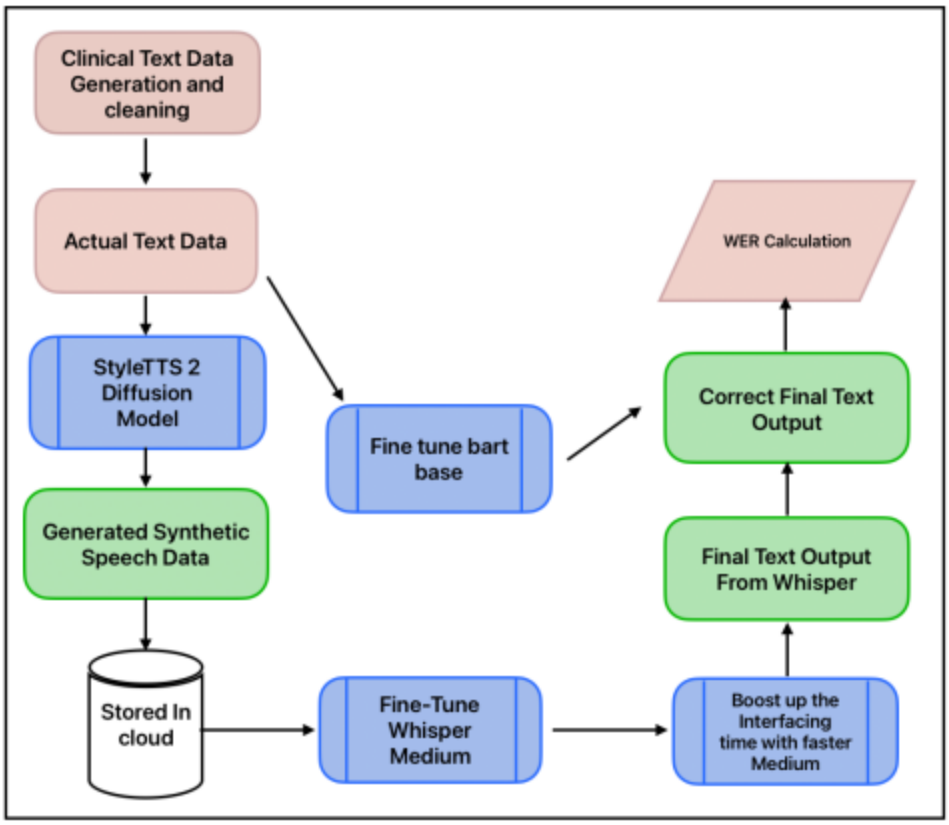


Figure 2.6, End-to-End Workflow of United-MedASR ASR System Development (Banerjee et al., 2024)

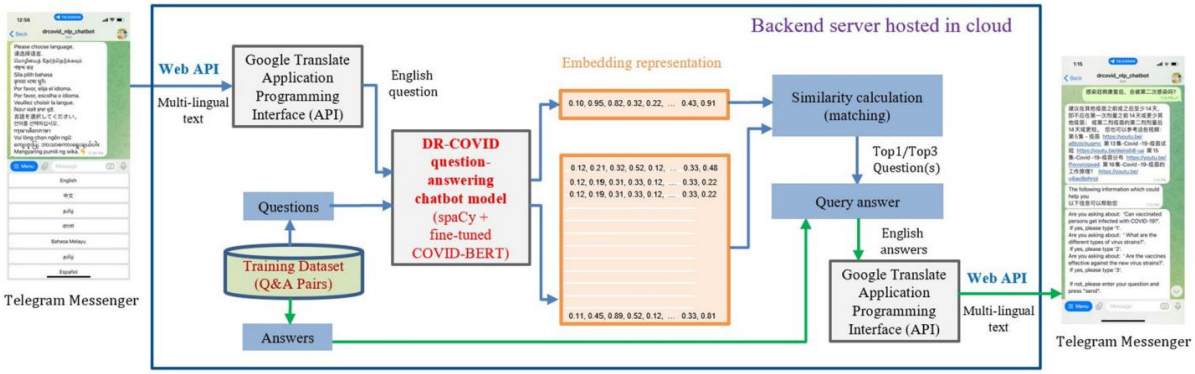


Figure 2.7, Overview of the DR-COVID NLP-based medical chatbot architecture (Yang et al., 2023)

2.5 Integration of Automatic Speech Recognition (ASR) and Semantic Modules

2.5.1 Challenges in Integrated Medical Chatbot Systems

Medical chatbot pipelines usually transcribe speech first and then apply downstream understanding modules, but ASR noise can still affect later tasks. Shapira et al. (2025) show that transcript errors propagate to downstream language understanding and can affect task performance differently depending on the error type.

This is important in medical dialogue because symptom text is often multi-label, long-tailed, and difficult to learn from when annotated data are limited. Mottaghi et al. (2020) report that one patient text can map to multiple symptoms and that rare symptoms are harder to recognize. Medical conversation models also need domain knowledge, since general generative models do not scale well in medical dialogue without medical-specific information. MKA improves medical conversation by injecting structured medical knowledge, and ChatDoctor also uses medical-domain knowledge and retrieval to improve response quality (Li et al., 2023; Liang et al., n.d.).

2.6 Research Gap

Although Whisper demonstrates robust speech-recognition performance, background noise, pronunciation differences, and recording quality can still produce transcription errors (Radford et al., 2022). These errors may alter important medical terms and propagate to downstream language-understanding tasks (Shapira et al., 2025).

Existing studies often evaluate speech recognition and medical text classification separately. Limited research directly examines audio preprocessing within an integrated medical chatbot while comparing traditional and transformer-based classifiers on the same dataset. Therefore, this study compares raw and pre-processed audio using WER and CER, and evaluates three text classification models which are TF-IDF with Logistic Regression, fine-tuned BERT, and fine-tuned BioBERT on the same dataset using accuracy, precision, recall, and F1-score. The model achieving the highest performance will be selected for integration into the speech-enabled medical chatbot. The selected classifier is integrated into a speech-enabled chatbot that provides preliminary health guidance rather than clinical diagnosis.

2.7 Summary

In summary, ASR errors can still affect downstream language understanding, while medical symptom text remains difficult to classify because it is multi-label and long-tailed (Shapira et al.,2025).

Therefore, an effective medical chatbot should combine speech-to-text conversion, fine-tuned disease classification , and safe response handling in a user-friendly prototype (Ke et al., n.d.; Mottaghi et al., 2021; Shapira et al., 2025; Shekar et al., 2024).

CHAPTER 3
RESEARCH METHODOLOGY

3.0 Introduction

This chapter describes the methodology of the proposed medical chatbot system. It covers the main phases of the project, beginning with speech-to-text conversion using Whisper, followed by text classification for symptom understanding, response generation for medical guidance, and prototype development for user interaction. The chapter also presents the tools and techniques used, the implementation steps, and the evaluation metrics used to measure system performance.

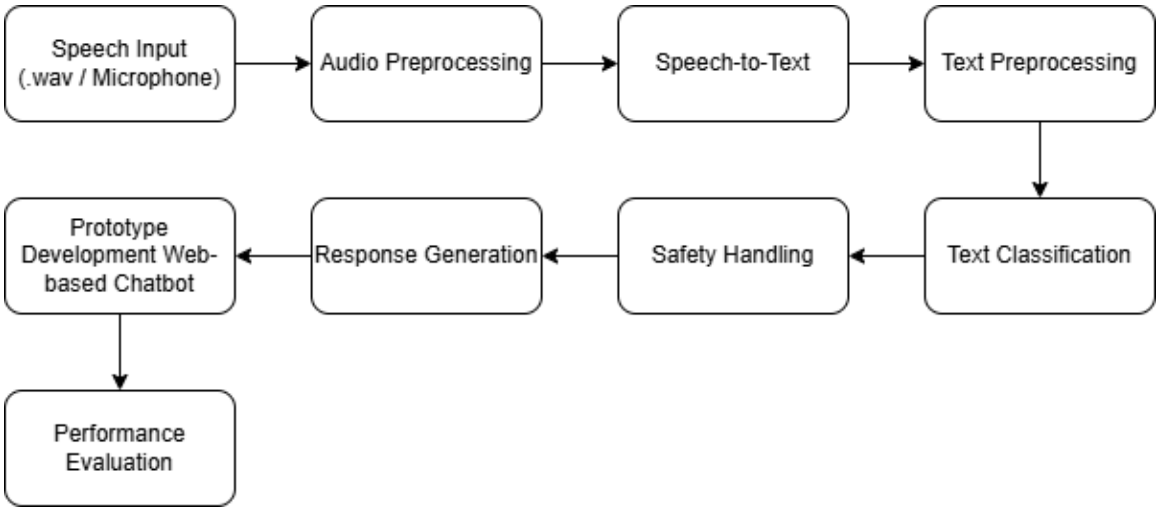


Figure 3.1, Overall Methodology Workflow

3.1 Speech to Text

This phase focuses on converting user speech into textual input using Automatic Speech Recognition (ASR). In this study, the Whisper model is utilized for speech-to-text conversion. The model is implemented locally during experimentation and evaluation, while Groq API may be used during prototype deployment for faster inference and real-time interaction.

Before transcription, audio preprocessing techniques are applied to improve input quality. These techniques include noise reduction, silence trimming, normalization, and resampling. The purpose of preprocessing is to reduce environmental noise and improve transcription accuracy before the audio is passed into the ASR model. Figure 3.1 shows the work flow of speech to text.

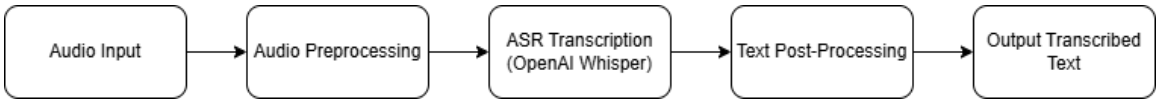


Figure 3.2, Speech to Text Workflow

Table 3.1, Speech-to-text Techniques

Module	Tools/ Technique	Function	Justification
Audio Input	Microphone/ Audio File	Capture user speech	Supports real-time speech interaction
Audio Preprocessing	Noise reduction, normalization	Improve audio quality	Reduce background noise before transcription
ASR	OpenAI Whisper	Convert speech to text	Robust multilingual ASR trained on large-scale data (Radford et al., 2022)

3.2 Text Classification

This phase focuses on predicting a disease class from textual symptom descriptions. Three text classification models are evaluated and compared on the same Symptom2Disease dataset: TF-IDF with Logistic Regression, fine-tuned BERT, and fine-tuned BioBERT. All

three models perform single-label classification across 24 disease classes and are evaluated using the same held-out test set. The model achieving the highest accuracy and weighted F1-score will be selected for integration into the chatbot pipeline. The predicted class supports preliminary health guidance and is not considered a clinical diagnosis.

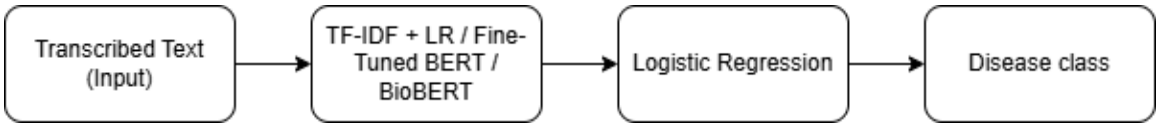


Figure 3.3, Text Classification Work Flow

Table 3.2, Text Classification Techniques

Module	Tools/ Technique	Function	Justification
Text Preprocessing	Lowercasing and TF-IDF vectorization using word unigrams and bigrams	Convert symptom descriptions into numerical features	Represents important words and short symptom phrases consistently
Text Classification	Logistic regression	Predict one of 24 disease classes	Achieved strong preliminary performance with low computational cost
Comparison model	Fine-tuned BERT	Provide a transformer-based comparison for the same classification task	Evaluates contextual classification against the traditional approach (Devlin et al., 2019)

	Fine-tuned BioBERT	Evaluate domain-specific transformer performance on symptom classification	Biomedical pre-training on PubMed corpora may improve recognition of medical terms compared to general BERT (Lee et al., 2020)
Dataset	Symptom-to-disease dataset (Kaggle)	Train and evaluate the classification models	Provides 1,200 labelled symptom descriptions across 24 disease classes

3.3 Response Generation and Safety Handling

After the disease class is predicted, the system generates a suitable medical response. The generated response may include basic guidance, follow-up recommendations, or warning messages depending on the predicted symptom severity. The response generation techniques are presented in Table 3.3.

To ensure safer interaction, the system includes an urgency detection mechanism. If severe symptoms are detected, the chatbot recommends the user to seek professional medical assistance. The system may also provide nearby clinic or hospital recommendations through a location module.

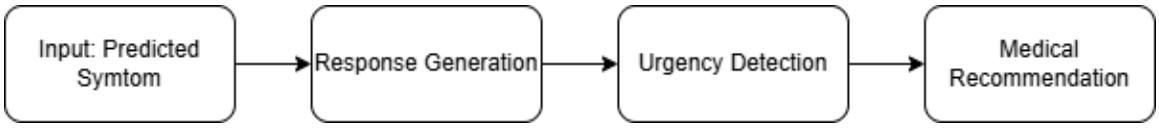


Figure 3.4, Response Generation and Safety Handling Work Flow

Table 3.3, Response Generation Techniques

Module	Tools/ Technique	Function	Justification
Response Generation	Rule-based / template-based response	Generate medical guidance	Ensures stable and consistent responses
Urgency Detection	Severity keyword/rule matching	Detect emergency symptoms	Supports safer medical recommendations
Location Recommendation	Geolocation / Maps API	Find nearby clinics/hospitals	Assists users during urgent situations

3.4 Baseline Models and Experimental Setup

This study compares the performance of audio preprocessing in two pipelines. For the baseline pipeline, raw audio is processed by Whisper directly without applying any pre-processing. For the improved pipeline, pre-processing steps such as DC-offset removal, trimming silence, stationary noise reduction and amplitude normalization are added before the pipeline is applied to raw audio for transcription. The audio preprocessing is the only independent variable for both pipelines, using the same Whisper model and TF-IDF with logistic regression classifier. The accuracy of the transcription is measured by Word Error Rate (WER) and Character Error Rate (CER). The robustness is also investigated with synthetic Gaussian noise at signal-to-noise ratios (SNRs) of 20, 10, 5, and 0 dB. The three models which are TF-IDF with Logistic Regression, fine-tuned BERT and fine-tuned BioBERT are evaluated on the same Symptom2Disease dataset for disease classification. The accuracy, macro and weighted precision, recall and F1-score metrics are used to evaluate all models. The model achieving the highest performance will be selected as the chatbot's primary classifier.

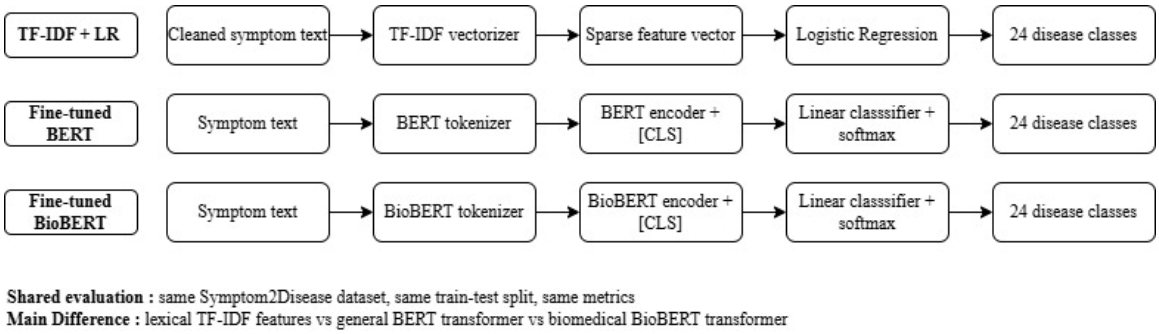


Figure 3.5, Architecture of three models

3.5 Evaluation Metrics

The proposed system is evaluated at two levels which are speech-to-text transcription and disease classification. Word Error Rate (WER) and Character Error Rate (CER) are used to evaluate Whisper transcription quality. Accuracy, precision, recall, and F1-score are used to evaluate all three classification models. (Radford et al., 2022).

3.5.1 Word Error Rate

Word Error Rate measures the difference between the reference transcript and Whisper's generated transcript at the word level:

$$WER = \frac{S + D + I}{N}$$

$$S = The\ number\ of\ word\ substitution$$

$$D = The\ number\ of\ word\ deletions$$

$$I = The\ number\ of\ word\ insertions$$

$$N = The\ total\ number\ of\ words\ in\ the\ reference\ transcript$$

A lower WER indicates better transcription performance.

3.5.2 Character Error Rate

$$CER = \frac{S_c + D_c + I_c}{N_c}$$

Where $S_c + D_c + I_c$ represent character substitutions, deletions, and insertions, while N_c is the number of characters in the reference transcript. A lower CER indicates better recognition of individual characters and medical terms.

3.5.3 Accuracy

Accuracy measures the proportion of test samples classified correctly:

$$Accuracy = \frac{Number\ of\ correct\ predictions}{Total\ number\ of\ predictions}$$

A higher accuracy indicates better overall disease-classification performance.

3.5.4 Precision

Precision measures how many samples predicted as a particular disease actually belong to that disease:

$$Precision = \frac{TP}{TP + FP}$$

where TP represents true positives and FP represents false positives.

3.5.5 Recall

Recall measures how many samples belonging to a disease are correctly identified:

$$Recall = \frac{TP}{TP + FN}$$

where FN represents false negatives.

3.5.6 F1-Score

The F1-score is the harmonic mean of precision and recall:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

Because the task contains 24 disease classes, weighted precision, weighted recall, and weighted F1-score are reported. Weighted metrics calculate each class's score and weight it according to the number of test samples in that class. Higher accuracy, precision, recall, and F1-score indicate better classification performance.

3.6 Prototype Development

In this phase, all modules are integrated into a medical chatbot prototype. The system interface allows users to interact through speech or text input. The chatbot displays transcription results, symptom predictions, and generated medical responses.

The prototype is developed using Python-based frameworks such as Flask or Streamlit. Optional image upload functionality may also be included for future extension of visual symptom analysis.

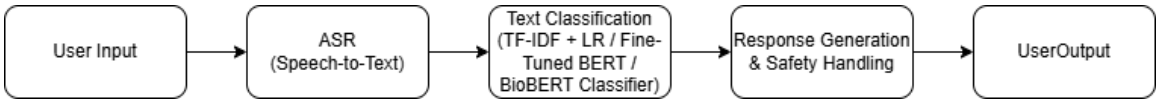


Figure 3.6, Prototype Development Work Flow

3.7 Gantt Chart and Milestones

The project schedule and major milestones are presented in the Appendix under the section titled "Gantt Chart and Milestones". The Gantt charts show the planned activities for Semester 202604 and Semester 202609, including proposal development, literature review,

dataset preparation, audio preprocessing, model comparison, chatbot prototype development, system testing, thesis writing, and final submission. The milestones table summarizes the key deliverables and expected completion period for each major stage of the project, allowing readers to understand the overall timeline and progress checkpoints.

CHAPTER 4
PRELIMINARY FINDINGS

4.0 Preliminary Results and Discussions

This chapter presents preliminary findings from two experiments. The first evaluates performance of Whisper transcription when audio is preprocessed in a real recording and synthetic Gaussian-noise environment. The second compares TF-IDF with Logistic Regression, fine-tuned BERT, and BioBERT on single-label classification across 24 disease classes. The three classifiers were evaluated against the same dataset of Symp2Disease and out-of-sample test set.

4.1 Audio Preprocessing Evaluation

The baseline and improved speech-processing pipelines are shown in Figure 4.1. Both configurations use the same Whisper model and downstream text classifier. The improved pipeline adds DC-offset removal, silence trimming, stationary noise reduction, and amplitude normalization before transcription. So, there is no difference, except audio pre-processing, between the two configurations.

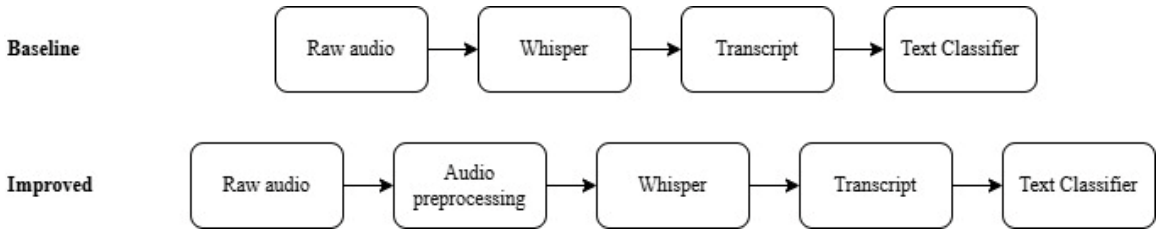


Figure 4.1, Baseline and improved speech-processing pipelines

4.1.1 Real Recording Comparison

The reference sentence for the real recording was "I have a headache every day in the morning." When the raw audio was processed directly, Whisper transcribed "headache" as "haydack", producing a Word Error Rate (WER) of 11.11% and a Character Error Rate (CER) of 9.52%. After audio preprocessing, the sentence was transcribed correctly, reducing both WER and CER to 0.00%, as shown in Table 4.1 and Figure 4.2.

Table 4.1, Whisper transcription results for the real recording

Pipeline	Whisper transcript	WER (%)	CER (%)
Baseline (raw audio)	I have a haydack every day in the morning.	11.11	9.52
Improved (preprocessed audio)	I have a headache every day in the morning.	0.00	0.00

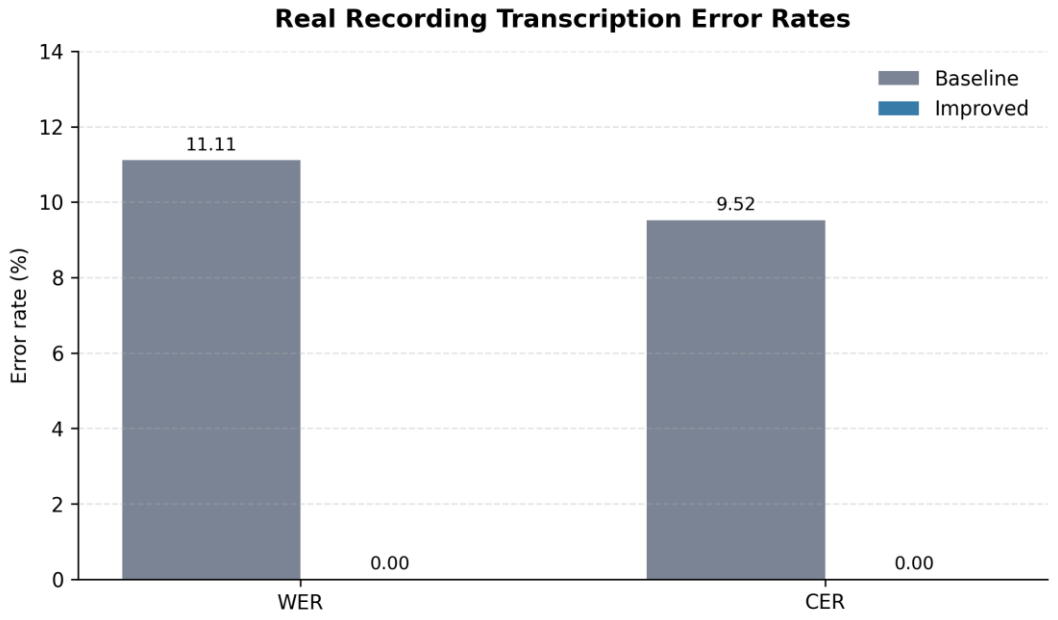


Figure 4.2, WER and CER comparison for the real recording

This recording did not contain a verified disease label. Consequently, it was used only to evaluate transcription quality and was not used to calculate disease-classification accuracy.

4.1.2 Synthetic-Noise Evaluation

Robustness was further evaluated by adding synthetic Gaussian noise at signal-to-noise ratios (SNRs) of 20, 10, 5, and 0 dB. Both pipelines produced the correct transcript from 20 to 5 dB. At 0 dB, where the noise level was most severe, preprocessing reduced WER from 33.33% to 22.22% and CER from 35.71% to 19.05%.

Table 4.2, Whisper performance under synthetic Gaussian noise

SNR (dB)	Baseline WER (%)	Improved WER (%)	Baseline CER (%)	Improved CER (%)
20	0.00	0.00	0.00	0.00
10	0.00	0.00	0.00	0.00
5	0.00	0.00	0.00	0.00
0	33.33	22.22	35.71	19.05
Mean	8.33	5.56	8.93	4.76

Across all four noise levels, the mean WER decreased from 8.33% to 5.56%, an improvement of 2.77 percentage points. Mean CER decreased from 8.93% to 4.76%, an improvement of 4.17 percentage points. These results indicate that preprocessing was most beneficial under the most challenging noise condition.

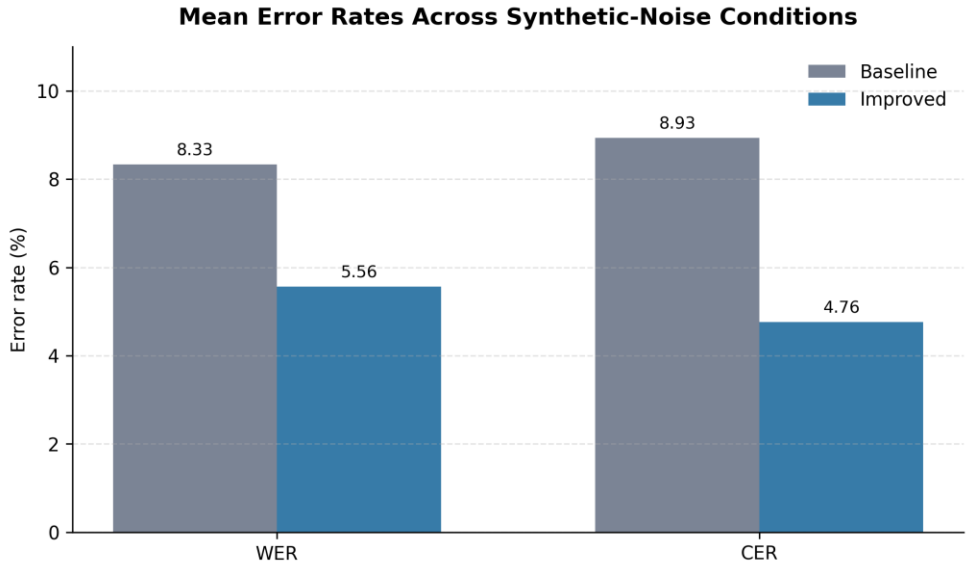


Figure 4.3, Mean WER and CER under synthetic Gaussian noise

4.2 Disease-Classification Evaluation

The text-classification experiment used the Symptom2Disease dataset, which contains 1,200 symptom descriptions distributed across 24 disease classes. A stratified split with a fixed random seed of 42 produced 960 training samples and 240 test samples. TF-IDF with Logistic Regression provides a sparse lexical baseline, fine-tuned BERT represents a general-purpose transformer, and BioBERT represents a biomedical-domain transformer. The same held-out test set and evaluation metrics were used for all three models.

6. Calculate identical metrics for all models

```
METRIC_COLUMNS = [
    "Accuracy",
    "Macro Precision",
    "Macro Recall",
    "Macro F1",
    "Weighted Precision",
    "Weighted Recall",
    "Weighted F1",
]


def calculate_metrics(model_name, labels, predictions):
    macro_precision, macro_recall, macro_f1, _ = (
        precision_recall_fscore_support(
            labels,
            predictions,
            average="macro",
            zero_division=0,
        )
    )
    weighted_precision, weighted_recall, weighted_f1, _ = (
        precision_recall_fscore_support(
            labels,
            predictions,
            average="weighted",
            zero_division=0,
        )
    )
```

```
    return {
        "Model": model_name,
        "Accuracy": accuracy_score(labels, predictions) * 100,
        "Macro Precision": macro_precision * 100,
        "Macro Recall": macro_recall * 100,
        "Macro F1": macro_f1 * 100,
        "Weighted Precision": weighted_precision * 100,
        "Weighted Recall": weighted_recall * 100,
        "Weighted F1": weighted_f1 * 100,
    }


comparison_df = pd.DataFrame(
    [
        calculate_metrics(model_name, y_true, predictions)
        for model_name, predictions in prediction_sets.items()
    ]
)
comparison_df[METRIC_COLUMNS] = comparison_df[METRIC_COLUMNS].round(2)

if not comparison_df[METRIC_COLUMNS].apply(
    lambda column: column.between(0, 100).all()
).all():
    raise AssertionError("All metric values must be between 0 and 100.")

ranked_comparison_df = comparison_df.sort_values(
    "Weighted F1",
    ascending=False,
).reset_index(drop=True)
ranked_comparison_df.insert(
    0,
    "Rank",
    np.arange(1, len(ranked_comparison_df) + 1),
)

best_by_metric = comparison_df[METRIC_COLUMNS].max()
difference_df = comparison_df[["Model"]].copy()
```

```python
for metric in METRIC_COLUMNS:
    difference_df[f"{metric} gap from best"] = (
        comparison_df[metric] - best_by_metric[metric]
    ).round(2)

display(ranked_comparison_df)
display(difference_df)
```

Figure 4.4, A sample of implemented code for calculating identical metrics for 3 models

Table 4.3, Classification performance of the three models (%)

Model	Acc.	Macro P	Macro R	Macro F1	Weighted P	Weighted R	Weighted F1
TF-IDF + LR	96.67	97.15	96.78	96.70	96.99	96.67	96.54
BioBERT	88.75	90.66	89.00	87.57	91.80	88.75	88.01
Fine-tuned BERT	87.92	88.27	87.62	86.94	88.58	87.92	87.25

TF-IDF with Logistic Regression achieved the highest result for every reported metric, including 96.67% accuracy and 96.54% weighted F1. BioBERT ranked second with 88.75% accuracy and 88.01% weighted F1, followed by fine-tuned BERT with 87.92% accuracy and 87.25% weighted F1. The strongest transformer result was therefore 8.53 percentage points below the TF-IDF model in weighted F1.

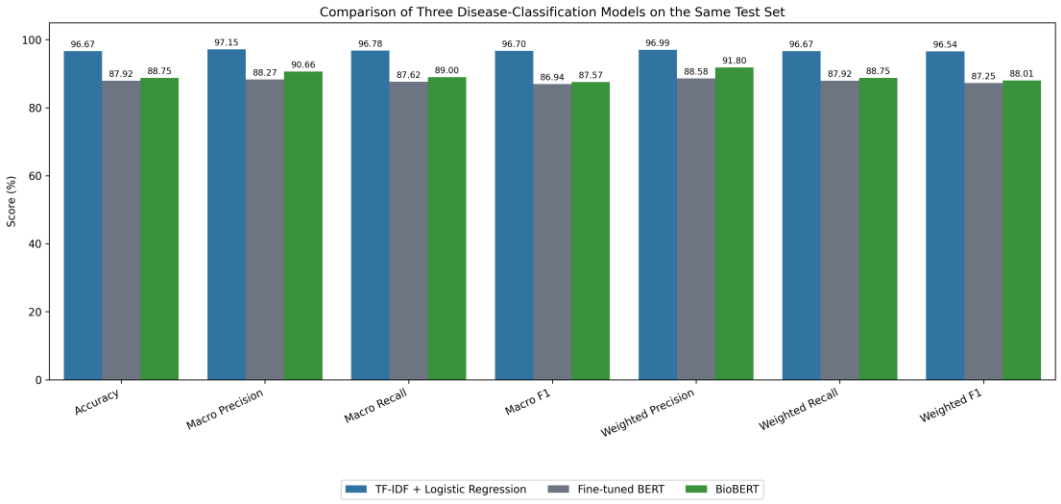


Figure 4.5, Performance comparison of TF-IDF + LR, BioBERT, and fine-tuned BERT

The result does not establish that TF-IDF with Logistic Regression is universally superior to transformer models. Its stronger performance is likely related to the small, balanced, and lexically structured dataset, where disease classes can be distinguished by recurring symptom terms. Transformer models may benefit more from larger and more diverse training data, domain-matched pretraining, and further hyperparameter optimization. These possibilities should be verified experimentally in future work.

4.3 Summary of Preliminary Findings

The preliminary audio experiment revealed an improvement of the pre-processing step when using Whisper to transcribe the real recording and a negative effect on the average error for the synthetic noise, especially at 0 dB. However, as the results are based on one source recording, the audio results should be confirmed with a larger sample of speakers, symptoms and recording environments in Thesis II.

For single-label, 24-class disease classification, TF-IDF with Logistic Regression produced the highest scores on the present dataset, while BioBERT slightly outperformed fine-tuned BERT. The three-model comparison will be used throughout the development of the medical chatbot with speech capabilities to assess accuracy, computational cost, latency, and scalability. Full pipeline integration and broader end-to-end evaluation will be conducted in Thesis II.

CHAPTER 5
CONCLUSION

5.0 Conclusion

This proposal presents a speech-enabled medical chatbot that integrates audio preprocessing, Whisper speech recognition, TF-IDF with logistic regression, and safety-based response generation. The system is used for single-label classification in 24 disease classes and will give the user some initial guidance in health but not diagnose a disease. Initial results show that audio preprocessing can decrease the number of transcription mistakes, especially in noisy scenarios for Whisper. TF-IDF with logistic regression also performed well on the selected dataset with an accuracy of 96.67% and a weighted F1-score of 96.54% with outperformance compared to the fine-tuned models of the BERT and BioBERT comparison models. These modules will be combined with a full medical chatbot in Thesis II and assessed based on other audio recordings, classification performance, handling of safety situations, quality of response, and usability.

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6

APPENDIX

GANTT CHART AND MILESTONES

Activity	Semester 202604													
	W1	**W2**	**W3**	**W4**	**W5**	**W6**	**W7**	**W8**	**W9**	**W10**	**W11**	**W12**	**W13**	**W14**
Proposal scope and research planning														
Literature review and research gap identification														
Dataset collection, inspection, and preprocessing														
Audio preprocessing and Whisper evaluation														
TF-IDF + LR, BERT, and BioBERT experiments														
Results analysis and three-model comparison														
Thesis proposal writing and revision														
Proposal presentation and submission														

Activity	Semester 202609													
	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14
System requirements analysis and architecture design														
Dataset expansion and data preparation														
Audio preprocessing and Whisper module refinement														
Classifier training, selection, and integration														
Chatbot backend and safety response development														
User interface and speech-input integration														
End-to-end testing and system evaluation														
Final thesis writing, revision, and demonstration														

MILESTONES		
Milestone	**Key deliverable(s)**	**Completion date**
Research foundation established	Research scope, objectives, literature review, and research gap completed	Semester 202604, W4
Preliminary speech pipeline completed	Audio preprocessing and Whisper WER/CER evaluation completed	Semester 202604, W7

MILESTONES		
Milestone	**Key deliverable(s)**	**Completion date**
Classification comparison completed	TF-IDF + LR, fine-tuned BERT, and BioBERT results compared	Semester 202604, W10
Thesis proposal completed	Revised proposal, presentation slides, and proposal submission completed	Semester 202604, W14
Medical chatbot prototype completed	Speech input, classifier, chatbot backend, safety responses, and interface integrated	Semester 202609, W9
Final evaluation and thesis completed	End-to-end testing, final thesis, system demonstration, and submission completed	Semester 202609, W14